

OpenFOAM-Based Comparative Study of Drag and Lift Behaviour in Bluff and Streamlined Flows

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Abstract—Fluid flow around solid bodies is a common phenomenon in everyday life, and understanding the resulting drag and lift forces is key to the performance and efficiency of many engineering systems. This study investigates incompressible, steady, laminar flow over three geometries — a circular cylinder, a square cylinder, and a thin flat plate — simulated using OpenFOAM, focusing on how velocity and viscosity influence drag and lift, and how this differs between bluff and streamlined bodies. In bluff bodies, pressure forces dominate, so drag is primarily due to pressure differences with minimal contribution from viscosity, while lift arises from pressure asymmetry between the upper and lower surfaces and is therefore more sensitive to viscosity. In streamlined bodies, pressure differences are minimal, so drag is caused mainly by viscous shear and responds directly to viscosity, while lift is comparatively unaffected.

Keywords—Incompressible flow; Steady flow; Circular cylinder; Square cylinder; Flat plate; Drag coefficient; Lift coefficient; OpenFOAM

I. INTRODUCTION

The generation of aerodynamic forces — drag and lift — is a fundamental aspect of external fluid flow, depending on a body's shape and orientation, the free-stream velocity and viscosity, and the fluid's density. Because it is impractical to tabulate these forces for every possible configuration, they are instead expressed as dimensionless coefficients — the drag coefficient (C_D) and lift coefficient (C_L) — allowing aerodynamic performance to be compared across a wide range of flow conditions.

This study investigates how velocity and viscosity individually affect drag, lift, and their coefficients across three fundamental geometries common in engineering — a circular cylinder, a square cylinder, and a thin flat plate — and examines how this behaviour differs between bluff and streamlined bodies.

Several studies have explored steady, incompressible flow over bluff bodies: Kravchenko and Moin^[3] and Kornbleuth^[2] each simulated a circular cylinder and established benchmark drag and lift data, while Xu et al.^[1] examined the effect of boundary constraint on flow around a square cylinder. Building on this benchmark work, the present study isolates the individual effects of velocity and viscosity within a fixed geometry, across thirteen cases spanning a circular cylinder, a square cylinder, and a flat plate, rather than reporting force values at a single operating point. It also contrasts a bluff body against a streamlined one; as discussed in the Results, this comparison is not fully controlled, since viscosity and length scale differ alongside geometry.

This distinction has practical value: in many engineering situations, only one of velocity or viscosity changes at a time — the same structure at different flow speeds, or the same shape in a different fluid. Knowing which parameter chiefly governs a given force offers a simpler basis for anticipating how it will change than assuming a fixed coefficient throughout. Bluff bodies such as cylinders are common in structures exposed to wind or water flow, while streamlined bodies such as plates and fins appear in control surfaces and heat-exchanger fins.

II. THEORETICAL BACKGROUND AND GOVERNING EQUATIONS

A fluid is considered incompressible when its density remains nearly constant throughout the flow, and laminar when the Reynolds number — the ratio of inertial to viscous forces — is below the critical value at which the flow transitions to turbulence, resulting in smooth, orderly motion. Viscosity is a fluid's internal resistance to shear; it gives rise to the no-slip

condition, whereby the fluid layer in contact with a stationary surface has zero velocity relative to it, producing a boundary layer — a thin region near the surface where velocity rises rapidly from zero at the wall to the free-stream value.

Flow over a bluff body — one with a blunt rear, such as a circular or square cylinder — forms a wake characterised by vortex shedding, causing asymmetric pressure distribution: the front-to-rear difference generates drag, and the top-to-bottom difference generates lift. Drag combines wall shear (friction drag) and pressure forces (pressure drag); in a streamlined body, such as a flat plate aligned with the flow, pressure differences are minimal, and drag is caused mainly by viscous shear.

Drag and lift are expressed as the dimensionless coefficients:

$$C_D = \frac{F_D}{\frac{1}{2}\rho V^2 A} \quad \dots (1) \qquad C_L = \frac{F_L}{\frac{1}{2}\rho V^2 A} \quad \dots (2)$$

where F_D and F_L are the computed drag and lift forces, ρ is the fluid density, V is the free-stream velocity, and A is the reference area used in calculating the coefficients. Density is treated as constant for these incompressible simulations, with $\rho = 1$ used in the simulation setup.

The flow is governed by the two-dimensional, steady, incompressible Navier–Stokes equations, with x aligned with the free-stream flow direction (along which drag acts) and y perpendicular to it (along which lift acts):

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \dots\dots (3)$$

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \rho f_x \quad \dots\dots(4)$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \rho f_y \quad \dots\dots(5)$$

Here u and v are the velocity components, p is pressure, and μ is dynamic viscosity; the time-derivative terms ($\partial u/\partial t$, $\partial v/\partial t$) and body-force terms (ρf_x , ρf_y) are all zero, as the flow is

steady and no external forces act on the fluid. Equation (3) enforces mass conservation; Equations (4) and (5) balance inertial forces against pressure gradients and viscous stresses, determining the pressure and velocity fields from which the drag and lift forces in Equations (1)–(2) are computed.

III. COMPUTATIONAL METHODOLOGY

A. Initial Conditions

The fluid used in this study is air. The steady-state simulations were run for up to 30,000 solver iterations, with results written every 100 iterations.

Three geometries were examined: a circular cylinder and a square cylinder as bluff bodies, and a flat plate as a streamlined body. Thirteen cases (Table 1) vary velocity V and kinematic viscosity ν independently for each geometry, as detailed in the Variable column.

The baseline case, A_1 , uses $V = 0.054$ m/s and $\nu = 1.5 \times 10^{-5}$ m²/s (approximating air), giving $Re = 180$. A real circular cylinder at this Reynolds number sheds vortices periodically — an unsteady phenomenon that the present steady, laminar formulation (Section III-C) does not resolve, consistent with the limitation noted in the Conclusion.

TABLE 1: SIMULATION CASES: GEOMETRY, FLOW PARAMETERS, AND REYNOLDS NUMBERS

Case	Shape	Domain Setup	Velocity (m/s)	Kinematic Viscosity (m ² /s)	Reynolds Number	Variable
A_1	Circular Cyl.	$D = 0.05$ m	0.054	1.5×10^{-5}	180	Baseline
B_1	Circular Cyl.	$D = 0.05$ m	0.054	7.5×10^{-6}	360	ν halved
C_1	Circular Cyl.	$D = 0.05$ m	0.216	1.5×10^{-5}	720	$V \times 4$
D_1	Circular Cyl.	$D = 0.05$ m	0.054	2.7×10^{-8}	1.0×10^5	ν very low
E_1	Circular Cyl.	$D = 0.05$ m	30.0	3.0×10^{-5}	5.0×10^4	High-Re baseline
F_1	Circular Cyl.	$D = 0.05$ m	30.0	1.5×10^{-5}	1.0×10^5	ν halved (ν s E_1)

G₁	Square Cyl.	L = 0.05 m	0.054	1.5×10^{-5}	180	Geometry change
A₂	Flat Plate	L = 1.0 m	0.054	7.5×10^{-5}	720	Baseline
B₂	Flat Plate	L = 1.0 m	0.054	3.75×10^{-5}	1440	ν halved
C₂	Flat Plate	L = 1.0 m	0.216	7.5×10^{-5}	2880	$V \times 4$
D₂	Flat Plate	L = 1.0 m	0.054	1.80×10^{-6}	3.0×10^4	ν very low
E₂	Flat Plate	L = 1.0 m	3.75	1.5×10^{-4}	2.5×10^4	High-Re baseline
F₂	Flat Plate	L = 1.0 m	3.75	7.5×10^{-5}	5.0×10^4	ν halved (ν s E ₂)

B. Assumptions

The flow is treated as incompressible, steady, and laminar — constant density, time-invariant properties, and turbulence-free motion — simplifying the governing equations to isolate the effects of shape, velocity, and viscosity on drag and lift.

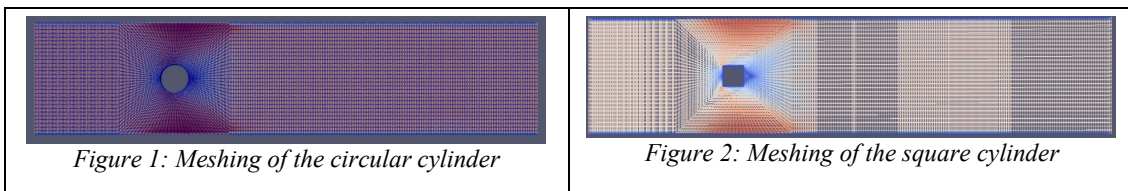
C. Solver

Simulations were carried out using simpleFOAM, a finite-volume OpenFOAM solver for steady-state, incompressible turbulent flow via the Reynolds-Averaged Navier–Stokes equations. The turbulence model was disabled here to simulate laminar conditions; in its absence, the solver solves the incompressible Navier–Stokes and continuity equations iteratively for the pressure and velocity fields.

Boundary conditions: fixed velocity with zero pressure gradient at the inlet; zero pressure and velocity gradients at the outlet; no-slip with zero pressure gradient at the walls.

D. Post-processing

ParaView was used for post-processing and visualisation of the simulated flow fields. Meshes for each geometry, refined near the body surface to resolve the boundary layer, are shown in Figures 1–3.



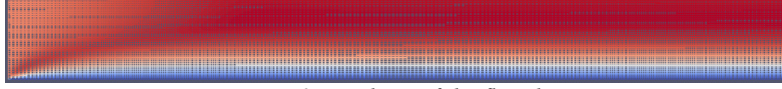


Figure 3: Meshing of the flat plate

Velocity Magnitude and Pressure Contours

Case A₁: Circular Cylinder with velocity 0.054m/s and kinematic viscosity $1.5 \times 10^{-5} \text{ m}^2/\text{s}$

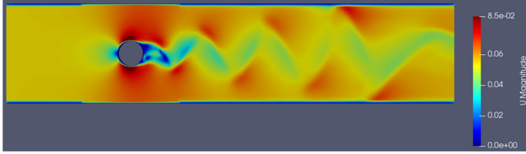


Figure 4: Velocity magnitude contour



Figure 5: Pressure contour

Case G₁: Square Cylinder with velocity 0.054m/s and kinematic viscosity $1.5 \times 10^{-5} \text{ m}^2/\text{s}$

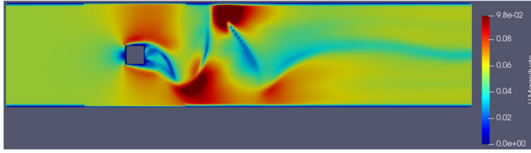


Figure 6: Velocity magnitude contour

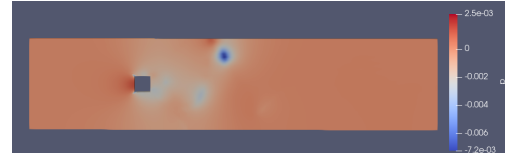


Figure 7: Pressure contour

Case A₂: Flat plate with velocity 0.054m/s and kinematic viscosity $7.5 \times 10^{-5} \text{ m}^2/\text{s}$

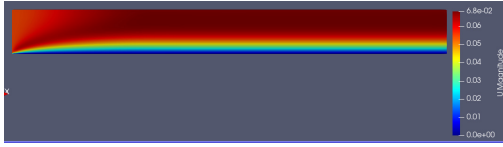
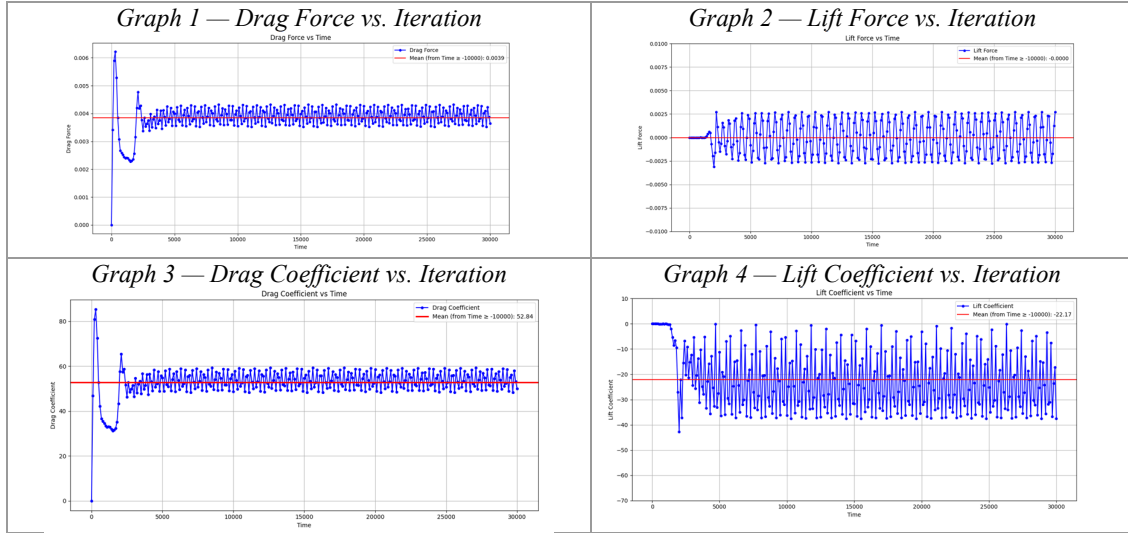


Figure 8: Velocity magnitude contour



Figure 9: Pressure contour

A force output file recorded pressure and viscous force components in the x, y and z directions at fixed iteration intervals for each simulation, allowing the convergence of the computed drag and lift forces to be monitored. To assess convergence, the iteration histories of drag force, lift force, and their coefficients were examined for representative Case A₁ (velocity 0.054 m/s, kinematic viscosity $1.5 \times 10^{-5} \text{ m}^2/\text{s}$), with a mean-value line added to each plot; small fluctuations about a stable mean indicated the solution had approached a steady numerical state, and the same check was applied to all other cases. Since simpleFOAM is a steady-state solver, these histories represent the solution's evolution over solver iterations, not physical time.



Graphs 1–4. Iteration Histories and Mean Values of Drag Force, Lift Force, Drag Coefficient, and Lift Coefficient for the Circular Cylinder with velocity 0.054m/s and viscosity $1.5 \times 10^{-5} \text{ m}^2/\text{s}$ (Case A1)

IV. RESULTS AND DISCUSSION

TABLE 2: PERCENTAGE CHANGE IN DRAG/LIFT FORCES AND COEFFICIENTS

Cases compared	% Change in Drag Force	% Change in Drag Coefficient	% Change in Lift Force	% Change in Lift Coefficient
Case A₁ and B₁	0.72%	0.7147%	38.35%	38.34%
Case A₁ and C₁	1405.83%	-5.891%	1468.63%	-2.018%
Case E₁ and F₁	10.13%	10.10%	16.40%	16.42%
Case A₁ and G₁	126.7%	126.8%	-62.69%	-62.69%
Case A₁ and A₂	-99.95%	-100%	-99.51%	-99.98%
Case A₂ and B₂	-41.36%	-41.36%	-36.01%	-36.02%
Case A₂ and C₂	468.1%	-64.5%	574.1%	-57.86%
Case E₂ and F₂	-33.51%	-33.51%	-28.76%	-28.76%

For the circular cylinder cases, where only velocity and viscosity were varied, the drag coefficient remained nearly constant — suggesting that for bluff bodies, the drag coefficient stays approximately constant within a given Reynolds-number range, until the overall flow characteristics change significantly. When velocity was increased four-fold (A₁ to C₁), drag rose by approximately 15 times (1405.83%), matching the near-quadratic relationship between drag and velocity given by Eq. (1).

In contrast, for a streamlined body like the flat plate, the drag coefficient decreases with increasing Reynolds number, so even when velocity was quadrupled, drag did not rise by the same proportion: the drop in the drag coefficient partly offset the effect of the increased velocity, so drag increased by only 468.1% (A₂ to C₂).

A. Bluff Bodies

In bluff bodies, drag is dominated by pressure differences from flow separation, with only a minor contribution from viscous shear. When viscosity was halved (A_1 to B_1), the resulting rise in Reynolds number caused only a small increase in drag (0.72%) but a much larger increase in lift (38.35%); since the present steady formulation does not resolve time-dependent wake dynamics such as vortex shedding, the mechanism behind this larger sensitivity cannot be established from these simulations alone. Since Reynolds number is inversely related to viscosity, lowering viscosity raises Re ; at higher Re , pressure forces dominate further and the influence of viscosity diminishes — consistent with the smaller lift change (16.40%) for the same halving of viscosity at higher Reynolds number (E_1 to F_1).

These results suggest that at low Reynolds numbers, lift in bluff-body flows is sensitive to viscosity through wake asymmetry, while drag remains comparatively stable, being governed mainly by pressure separation. Changing geometry from circular to square cylinder (A_1 to G_1), with velocity, viscosity and length scale unchanged, increased drag by 126.7% and reduced lift by 62.69%.

Because the cylinders are simulated at zero incidence with symmetric geometry, the underlying lift force is small in absolute terms compared to drag; the percentage changes reported above should therefore be read as the sensitivity of a small quantity, rather than as evidence of a large physical lift force.

B. Streamlined Bodies

In streamlined bodies such as the flat plate, viscous shear is the primary contributor to drag, especially at low Reynolds numbers where the flow remains attached. At the same velocity, the flat plate produced substantially lower drag and lift than the circular cylinder (A_1 vs A_2 : -99.95% and -99.51% respectively), consistent with its streamlined profile; this comparison, however, also changes viscosity and length scale alongside geometry, so it reflects the

combined effect of shape, viscosity and size rather than shape alone — unlike the circular-to-square comparison (A_1 vs G_1) discussed above, which isolates geometry cleanly.

When viscosity was reduced (A_2 to B_2), drag decreased by 41.36% and lift by 36.01%, showing drag to be more sensitive to viscosity here; since a flat plate aligned with the flow produces minimal pressure-induced lift, lift is less affected by viscosity changes.

Reynolds number affects the size of this viscosity sensitivity: the drag reduction from halving viscosity was larger at low Re (41.36%, A_2 to B_2) than at high Re (33.51%, E_2 to F_2). This is consistent with the boundary layer being comparatively thicker, and less fully developed, at the lower Reynolds numbers simulated here; as Re increases and the boundary layer thins relative to the plate length, its response to a given change in viscosity becomes more consistent. Pressure forces remain minimal for this streamlined geometry throughout, as described in Section II.

For both bluff and streamlined bodies, drag and lift coefficients are obtained by normalising forces by $\rho V^2 A$. When viscosity alone changes, this denominator is unchanged, so the coefficient tracks the force directly, and the two change by similar amounts, as seen throughout Table 2. When velocity changes, both the force and the V^2 term change together; since force typically grows more slowly than V^2 , the coefficients decrease even as the raw forces increase.

V. CONCLUSION

For bluff bodies, the drag and lift coefficients remained nearly constant across the velocity and viscosity changes tested, indicating that within this Reynolds-number range they are governed primarily by geometry. Quadrupling velocity increased drag and lift by approximately 15 times, matching the quadratic velocity dependence in the drag equation; lift was more sensitive than drag to viscosity, particularly at low Reynolds numbers, though —

since the steady formulation cannot resolve vortex shedding — this should be read with caution rather than attributed to a specific mechanism.

For the flat plate, coefficients varied more with Reynolds number, since boundary-layer development and separation are Re-sensitive; drag and lift therefore did not scale simply with velocity, and viscosity affected drag more than lift. Changing the cylinder cross-section from circular to square, at matched velocity, viscosity and length scale, increased drag by 126.7% and reduced lift by 62.69%, cleanly isolating geometry's effect; the flat plate showed still lower forces, though that comparison also involves simultaneous changes in viscosity and length scale.

While drag and lift coefficients are dimensionless, they are functions of the Reynolds number, which encapsulates the combined effects of velocity, viscosity and density; changes in these parameters alter the flow structure — the separation point and pressure distribution — and hence the coefficients, explaining the variation observed here despite their non-dimensional formulation.

These findings are relevant wherever a design must perform across a range of flow speeds or fluids: since, as shown here, velocity and viscosity affect bluff and streamlined bodies differently, assuming a single fixed coefficient across operating conditions can be misleading.

It should be noted that the laminar assumption here is a modelling choice — made by disabling the turbulence model — rather than a description of the physical flow regime; at the higher Reynolds numbers simulated (up to 1.0×10^5), real cylinder flow is turbulent and unsteady, so results at these conditions should be read as trends within this laminar model, not as predictions of the actual physical flow.

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